

CHAPEL HILL, NC, USA

WMO: 746939

Lat: 35.933N

Lon: 79.064W

Elev: 512

StdP: 14.43

Time Zone: -5.00 (NAE)

Period: 01-18

WBAN: 93785

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	23.5	-0.2	5.5	25.4	4.4	7.0	29.3	17.0	38.1	15.7	37.9	4.1	300	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	94.3	74.4	91.9	74.1	89.9	73.5	77.5	89.0	76.5	87.1	75.5	85.4	5.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	130.9	81.3	73.3	126.3	80.5	72.6	123.3	79.8	41.1	89.0	40.1	87.3	39.2	85.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.7	12.2	10.7	DB	13.8	98.2	4.0	3.3	10.9	100.6	8.6	102.5	6.3	104.3	3.4	106.7
			WB	11.4	79.0	3.7	1.0	8.8	79.8	6.6	80.4	4.5	80.9	1.8	81.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	40.6	43.6	51.4	60.7	67.6	75.9	79.2	78.2	71.9	61.1	51.4	44.8
	DBStd	15.50	10.17	9.48	9.90	8.26	6.95	4.94	3.97	4.40	5.58	7.79	8.28	9.16
	HDD50	972	324	224	107	12	0	0	0	0	0	10	79	216
	HDD65	3248	757	600	434	179	58	1	0	0	11	169	412	627
	CDD50	4851	32	46	149	332	546	777	905	874	658	354	122	56
	CDD65	1652	0	2	11	48	139	328	440	409	220	48	5	2
	CDH74	14641	2	8	96	477	1160	3038	4184	3703	1600	342	28	3
	CDH80	5842	0	0	7	105	337	1278	1888	1615	548	63	1	0
Wind	WSAvg	3.8	4.2	4.5	4.8	4.5	3.9	3.4	2.9	3.0	3.8	3.7	3.7	3.8
Precipitation	PrecAvg	46.7	4.0	3.7	4.4	3.3	4.1	4.1	4.3	4.6	3.9	3.5	3.4	3.3
	PrecMax	61.3	8.9	6.8	8.0	7.4	7.6	10.2	8.4	9.1	24.0	10.1	12.6	7.2
	PrecMin	33.3	1.2	0.6	1.5	0.7	1.1	0.9	0.3	0.8	0.1	0.3	0.6	0.3
	PrecStd	7.1	1.9	1.6	1.9	1.6	1.7	2.2	2.0	1.9	3.9	2.3	2.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.5	74.6	81.0	87.5	89.3	95.4	98.4	97.9	92.9	85.2	77.4	73.3
		MCWB	60.6	62.6	62.2	66.9	71.0	73.9	75.4	74.5	70.2	68.5	64.2	62.1
	2%	DB	65.7	69.6	76.7	82.9	86.5	92.6	94.3	94.0	89.3	81.4	73.2	69.1
		MCWB	58.0	57.4	60.5	64.8	69.7	73.5	75.6	74.3	71.4	67.7	61.6	62.3
	5%	DB	62.0	65.0	72.6	79.7	84.0	90.1	91.7	91.0	86.2	77.9	69.9	64.5
		MCWB	54.9	55.5	58.5	63.2	69.3	73.0	75.2	73.9	70.2	66.1	61.5	59.0
	10%	DB	57.2	60.4	68.0	75.9	81.0	87.6	89.5	88.5	82.6	74.3	66.0	60.7
		MCWB	50.3	51.0	56.3	62.0	67.6	72.1	74.6	73.3	69.4	65.2	58.7	54.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.8	64.9	65.4	71.5	74.5	77.4	78.9	78.8	76.2	75.0	69.2	66.6
		MCDB	67.4	72.1	75.0	80.2	84.8	89.2	91.9	89.2	84.9	80.1	70.9	70.4
	2%	WB	60.0	61.3	63.4	68.3	72.0	76.0	77.7	77.4	74.8	71.9	65.6	63.4
		MCDB	64.4	66.8	71.5	77.7	82.2	87.3	90.2	87.8	82.6	76.4	70.2	66.9
	5%	WB	56.6	57.4	61.5	65.9	70.6	75.0	76.8	76.3	73.3	69.3	63.2	60.2
		MCDB	60.8	63.4	69.9	75.2	80.3	85.7	88.3	86.0	80.2	74.7	67.7	64.8
	10%	WB	51.3	51.9	59.1	64.0	69.2	73.9	75.8	75.2	72.0	66.6	59.8	55.2
		MCDB	56.0	58.0	66.3	72.6	77.8	83.8	86.2	84.2	78.2	72.6	65.3	59.3
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.7	21.3	22.4	20.3	20.0	18.9	18.6	18.5	20.5	21.6	18.5
		MCDBR	23.2	24.2	24.8	25.9	23.0	22.9	21.9	22.4	23.1	23.3	24.0	21.8
		MCWBR	16.5	15.6	12.9	11.5	9.8	8.2	7.2	6.8	8.2	11.3	14.5	15.3
	5% WB	MCDBR	20.5	22.0	21.5	21.5	19.7	19.8	19.3	18.3	16.7	17.1	19.0	18.8
		MCWBR	16.9	16.9	13.9	11.5	9.2	8.0	7.1	6.8	7.8	10.1	14.1	15.0
Clear-Sky Solar Irradiance	taub	0.312	0.330	0.363	0.404	0.451	0.497	0.539	0.528	0.422	0.380	0.330	0.315	
	taud	2.501	2.444	2.390	2.274	2.175	2.089	1.989	2.005	2.299	2.392	2.494	2.526	
	Ebn at Noon	282	289	287	279	266	253	241	241	264	266	272	273	
	Edh at Noon	26	31	36	42	47	51	56	55	39	32	26	24	
All-Sky Solar Radiation	RadAvg	811	1026	1355	1704	1886	2023	1902	1726	1438	1168	890	700	
	RadStd	54	122	96	127	150	143	118	87	127	150	70	59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+263	+178

Nomenclature: See separate page